Question no.1
Find standard deviation for the following data
36, 42, 47, 29, 41, 45, 52, 37, 59, 63, 29

Solution:
Mean of the data = 43.63

x	x - 43.63	(x -43.27)^2
36	-7.63	58.2169
42	-1.63	2.6569
47	3.37	11.3569
29	-14.63	214.0369
41	-2.63	6.9169
45	1.37	1.8769
52	8.37	70.0569
37	-6.63	43.9569
59	15.37	236.2369
63	19.37	375.1969
29	-14.63	214.0369
Total		**1234.5459**

Now variance = 1234.5459/(11) = 112.2314
Standard Deviation = 10.5939

Question no.2
Find Mean absolute Deviation from Mean of 3, 4 and 2with mean = 3.

Solution:

X	$\left\|X-\overline{X}\right\|$
3	0
4	1
2	1
Total	2

Mean =4

$$M.D=\frac{\sum\left|X-\overline{X}\right|}{n}$$

$$=\frac{2}{3}$$

Question no.3

What is the essential information which must be given to give a five number summary of the data set?

Solution:

First quartile
Median
Third quartile
Lowest value
Highest value

Question no.4
Average sale and standard deviation for a store are 18 and 6.5 respectively .Find coefficient of variation.

Solution:

$$C.V = \left(\frac{S}{\bar{x}}\right) * 100$$

$$= \left(\frac{6.5}{18}\right) * 100$$

$$= 36.11$$

Question no.5
Mean and median of frequency distributions are 18 and 25respectively. Find mode of the distribution.

Solution:

Mode= 3 median – 2 mean

Mode = 3 (25) –2(18)
= 75-36=39

Question no.6
An automobile dealership pays to its salespeople, a salary plus a commission on sales. The mean biweekly commission is $1395, the median is $1350 and the mode is $65.

Is the distribution of commissions positively skewed, negatively skewed, or symmetrical? Give reason.

Solution:

Since,
Mean > Median >Mode
Hence the distribution of commission is positively skewed.

Question no.7
A bag contains 9 balls. Among them 4 are blue and 5 are red. One ball is taken out at random from the bag. Find the probability the ball is red.

Solution:

Total number of balls = 9

Probability that the first ball drawn is blue = 4/9.